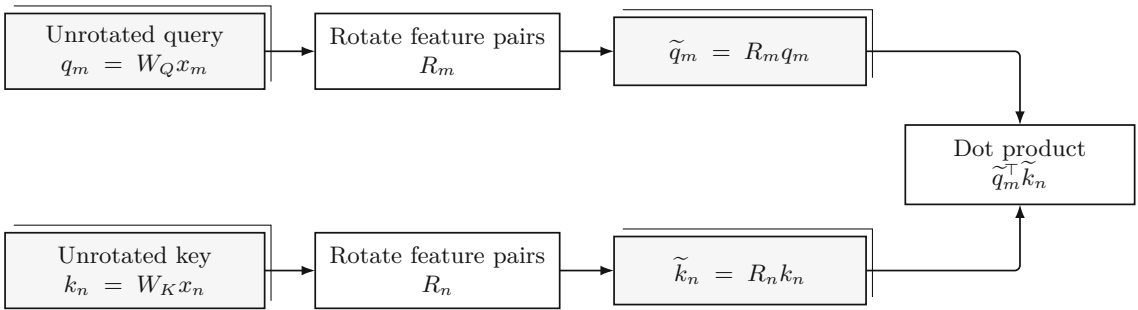
Rotary positions act on query and key feature pairs

Conceptual schematic; symbolic dimensions; batch axis omitted



$$\text{Each feature-pair block: } R_m^{(a)} = \begin{bmatrix} \cos(m\theta_a) & -\sin(m\theta_a) \\ \sin(m\theta_a) & \cos(m\theta_a) \end{bmatrix}$$

$$\widetilde{q}_m^\top \widetilde{k}_n = q_m^\top R_{n-m} k_n$$

Position rotations change query–key dot products through relative position; values follow their ordinary projection path.